

Course Introduction: AI and Applied Economics

Summer School in AI & Applied Economics

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ETH Zürich

August 2026

Outline

Motivation: Why AI for Economists

Course Overview and Roadmap

A Brief Tour: Text as Data

LLMs in Economic Research

The Frontier: What Comes Next

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Welcome

- ▶ This course is about using **artificial intelligence** as a working tool in **applied economics and social science**.
- ▶ **Engineering goals:**
 - ▶ develop practical skills in machine learning, natural language processing, and agentic workflows;
 - ▶ apply them to real economic data — legal, political, media, and firm-level text and images.
- ▶ **Scientific goals:**
 - ▶ relate unstructured data to metadata in order to measure social and economic forces;
 - ▶ understand the decisions of judges, legislators, journalists, and firms through what they write and say.
- ▶ **Policy goals:**
 - ▶ ask what happens to labor markets, law, and science when machines can read and write.

Why now: the cost of reading collapsed

- ▶ Economists have always wanted to use documents as evidence. Until recently, reading them did not scale.
- ▶ Three things changed:
 1. **Supply of digitized data:** court opinions, legislative records, patents, filings, newspapers, parliamentary transcripts, satellite and street-level imagery.
 2. **Methods:** representation learning, transformers, instruction-tuned language models.
 3. **Cost:** annotating a million documents went from a research grant to an afternoon.

The economist's version of the claim

AI does not (yet) answer economic questions. It dramatically lowers the cost of *constructing the variables* that economic questions require.

Three waves of text and AI in economics

1. **Dictionaries and bag-of-words** (roughly 2005–2015)
 - ▶ counts of chosen words; transparent, cheap, easily criticized;
 - ▶ e.g., Baker, Bloom, and Davis (QJE 2016) economic policy uncertainty.
 2. **Learned representations** (roughly 2013–2019)
 - ▶ word2vec, GloVe, topic models, then contextual embeddings (BERT);
 - ▶ measurement of latent dimensions: ideology, slant, sentiment, culture.
 3. **Large language models and agents** (2020–)
 - ▶ zero-shot annotation, extraction, summarization, and now code execution and multi-step research workflows.
- ▶ Each wave lowered the cost of *proposing* a measure. None removed the obligation to *validate* it.

Two distinct kinds of gains

- ▶ **Speed-ups:** things we already did, done faster and cheaper.
 - ▶ cleaning and linking messy administrative data;
 - ▶ coding a few thousand documents an RA would have coded;
 - ▶ writing and debugging estimation code;
 - ▶ drafting, translating, and literature search.
- ▶ **New methods:** research designs that were previously infeasible.
 - ▶ measuring dimensions with no closed-form dictionary (tone toward a group, legal ambiguity, argument structure);
 - ▶ corpus-scale designs: every judicial opinion, every parliamentary speech, every local newspaper;
 - ▶ automated hypothesis generation, simulation, and replication.

A useful discipline

For every application: is this a speed-up, or a new method? The econometric issues are different.

The central caution: measurement error with structure

- ▶ An AI-generated variable $\hat{Y}$ is a *prediction* of a latent construct Y^* , not the construct itself.
- ▶ The error $\hat{Y} - Y^*$ is:
 - ▶ **not classical** — it correlates with the text, and hence with the treatment and the outcome;
 - ▶ **systematic across documents** — the same model makes the same mistake everywhere in your panel;
 - ▶ **invisible** — cheap to produce, expensive to audit.
- ▶ Consequences: attenuation, spurious significance, and confounding by writing style rather than substance.
- ▶ We return to this in the econometrics lecture; treat it as the running theme of the whole Summer School.

Example: text reveals things metadata cannot

Ash, Goessmann, and MacLeod (2022) — judicial writing style and age

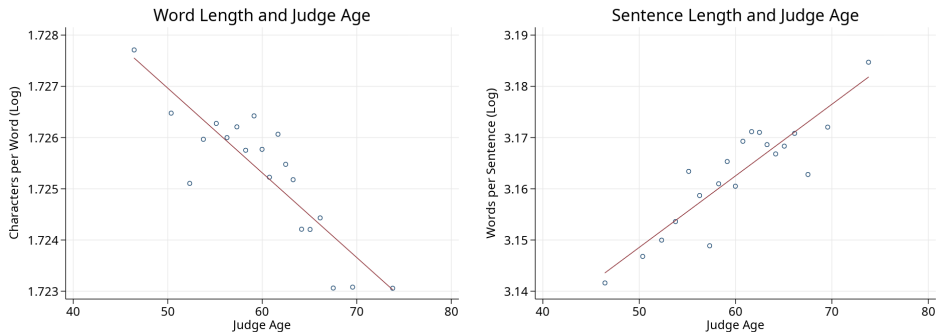


Figure 1: Average word length and sentence length in U.S. judicial opinions, by judge age. Style features are measurable at corpus scale and link directly to individual-level metadata.

Documents sit inside a system, and the system is text

- ▶ **Institutions:** constitutions, charters, treaties
- ▶ **Elections and policymaking:** campaign ads, parliamentary debates, proposed bills
- ▶ **Media:** newspapers, TV transcripts, lobbying, academic research
- ▶ **Laws and policies:** legislation, regulation, judicial opinions
- ▶ **Outcomes:** contracts, culture, prices

Every node is observable as text. That is what makes the whole chain newly measurable.

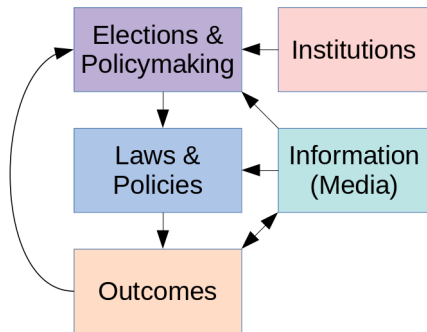


Figure 2: Text-producing components of a political-legal system.

Scale of the legal corpus alone

Ash, Morelli, and Vannoni (2022)

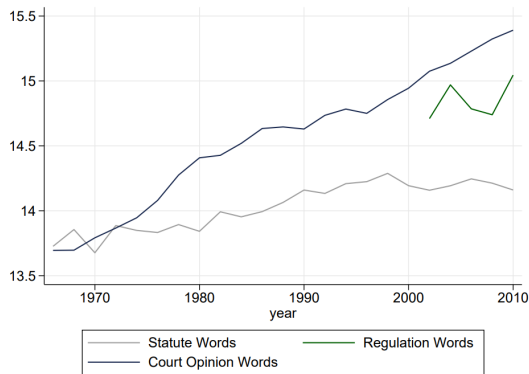


Figure 3: Annual legal text output in U.S. states (log scale).

Per year, U.S. states produce roughly:

- ▶ ~1.3M words of statutes
- ▶ ~3.3M words of regulations
- ▶ ~4.8M words of state court opinions

No research team reads this.
A model can.

AI is itself an object of economic study



Figure 4: AI adoption and innovation.



Figure 5: Measured productivity effects.

Noy and Zhang (2023): generative AI compresses the writing-quality distribution, with the largest gains for the lowest-skilled. Brynjolfsson, Li, and Raymond (2025): the same pattern for customer support agents in the field.

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Structure of the Summer School

- ▶ **Week 1 — online (Zoom), 31 August – 4 September, afternoons CET**
 - ▶ eight lectures over four afternoons, plus a Friday recap-and-presentations session;
 - ▶ lectures pair concepts with hands-on notebooks.
- ▶ **Week 2 — in person at ETH Zürich (RZ building, room F21), 7–10 September**
 - ▶ faculty research sessions each morning;
 - ▶ project studio and advisor meetings each afternoon.
- ▶ **Organizers:** Elliott Ash, Sergio Galletta, Joachim Voth, David Yanagizawa-Drott.
Instructors: Andrea Ciccarone, Felix Drinkall, Germain Gauthier, Fabian Roeben.
Advisors: Felix Chopra, Carlo Schwarz.

Week 1 roadmap (1/2)

Monday

1. **Course introduction** — AI and applied economics (*today, Ash*)
2. **ML basics, embeddings, and clustering** — prediction, regularization, validation; representations and unsupervised structure (*Ciccarone*)
3. **Text and image data** — text and images as economic measurement (*Ciccarone*)

Tuesday

4. **ML and econometrics** — inference with machine-generated variables (*Gauthier*)
5. **LLMs and alignment** — how a language model works (tokens, attention, transformers), then instruction tuning, RLHF, reasoning, RAG, and agents (*Ash*)

Week 1 roadmap (2/2)

Wednesday

- 6. **Working with AI agents** — orchestration, tooling, evaluation (*Roeben*)
- 7. **Software and application development with AI** — building research infrastructure (*Roeben*)

Thursday

- 8. **Training LLMs** — running and adapting models in practice: APIs, open weights, fine-tuning, LoRA (*Drinkall*)

Friday: recap, conclusion, and participant presentations (*Ash*)

- The arc: *measure* $\rightarrow$ *model* $\rightarrow$ *infer* $\rightarrow$ *automate*.

Week 2: research sessions in Zürich

- ▶ **Mon:** Sergio Galletta — ML for anti-corruption policy; bots as experimental treatments (BallotBot).
- ▶ **Tue:** Elliott Ash — worker rights in collective bargaining; generative AI in courts. Philine Widmer — deep latent variable models for social science (Chorale).
- ▶ **Wed:** Joachim Voth — revealing life preferences through LLMs; visual bias; time-locked LLMs.
- ▶ **Thu:** David Yanagizawa-Drott — generative AI in a teacher hiring pipeline; autonomous policy evaluation.
- ▶ Afternoons: **project studio**, advisor meetings, consultations.

What we want you to notice

In good applied AI papers, the model is usually the least interesting part. The design, the validation, and the identification carry the paper.

Assignments

- ▶ **Week 1: problem set.**

- ▶ based on the methods and workflows introduced during the online week; scope and deadline announced during the week.

- ▶ **Week 2: research idea.**

- ▶ a research idea that uses AI as a *method*, an *object of study*, or a *tool for building a research application*;
- ▶ developed in the project studio sessions with advisor feedback;
- ▶ a pilot on a small sample is worth more than a grand design — ideas from this course should be able to become working papers.

Preparation and tools

- ▶ Bring a **laptop** suitable for coding, set up before Week 1:
 1. access to a capable **AI service** — ChatGPT Plus or Claude Pro (or higher); contact the organizers if this is not feasible;
 2. at least one **coding agent** installed: Codex and/or Claude Code;
 3. **Git** and a working **Python/Jupyter** environment.
- ▶ Working libraries: `scikit-learn`, `pytorch`, `transformers`, `datasets`, `spacy`, `bertopic`.
- ▶ Notebooks and data links are distributed in the course repository.
- ▶ Corpora: get familiar with Hugging Face Datasets (<https://huggingface.co/docs/datasets/>), REST APIs, and basic scraping — not assessed, but the binding constraint on most projects.

How to get the most out of this course

- ▶ **Bring a question, not a method.** The best projects here start from an economic question that a corpus can answer.
- ▶ **Run the notebooks during the lecture.** The friction is the lesson.
- ▶ **Break the models on purpose.** Feed them adversarial documents and see where the measure fails.
- ▶ **Validate everything you measure** against human labels, even a hand-coded sample of 100 documents.
- ▶ **Use the agents openly.** We assume you will; we care whether you can tell when they are wrong.

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What is the endpoint of NLP?

Machine understanding of **discourse** across long documents and corpora.

- ▶ good summaries of long texts: extract the relevant, discard the irrelevant;
 - ▶ question answering: retrieve evidence and answers from large corpora;
 - ▶ reasoning over a body of documents that no human could read.
-
- ▶ Are we there now?
 - ▶ If not, what is missing — capability, evaluation, or data?

Four modes for NLP

- ▶ **Local**: linguistic information from local context (sentences, paragraphs)
 - ▶ local sentiment, textual entailment, co-reference resolution, closed question answering.
- ▶ **Long document** (covering multiple topics)
 - ▶ TF-IDF and bag-of-words representations → supervised learning; cosine distance between document vectors.
- ▶ **Global / knowledge base**: corpus-level tasks
 - ▶ information retrieval and search; open question answering; claim checking.
- ▶ **Generative / creative**: produce text for a purpose
 - ▶ draft a summary, a legal brief, a counterargument, or a simulated survey response.

Objectives: social-science research using text data

1. **What is the research question?**
2. **Corpus and data**
 - ▶ obtain, clean, preprocess, and link to metadata;
 - ▶ produce descriptive statistics and visuals on text and metadata.
3. **Language modeling**
 - ▶ **what exactly are we trying to measure?**
 - ▶ select a model and train (or prompt) it;
 - ▶ probe sensitivity to hyperparameters and prompts;
 - ▶ **validate** that it measures what you claim.
4. **Empirical analysis**
 - ▶ produce statistics or predictions with the trained model;
 - ▶ **answer the research question.**

Text is high-dimensional

- ▶ Text data is a sequence of characters called **documents**; the set of documents is the **corpus** D .
- ▶ For documents of fixed length n_L drawn from a vocabulary of size n_V , the number of distinct possible documents is $n_V^{n_L}$.
 - ▶ 30-word messages using only the 1,000 most common English words:
 $(10^3)^{30} = 10^{90}$ distinct documents.
- ▶ Represent a document as a matrix of n_V -dimensional categorical variables:

$$\mathbf{X} = [\mathbf{x}_1 \quad \cdots \quad \mathbf{x}_t \quad \cdots \quad \mathbf{x}_{n_L}]$$

- ▶ dimensionality of one document = $n_L n_V$, sparse with n_L non-zero entries;
- ▶ for $n_L = 30$, $n_V = 1,000$: dimensionality = 30,000.

Text data is unstructured

- ▶ The information we want is mixed together with (a lot of) information we do not want.
- ▶ **Every** text-as-data approach throws away information. The trick is choosing what to keep.
- ▶ Tokenization and dimension reduction are exactly this step: transforming an unstructured corpus D into a usable matrix X .

The same logic applies to LLMs

Prompting a model to output a number is also a dimension-reduction choice — it is just made implicitly, by the model, rather than explicitly, by you.

Local structure matters: co-reference

The legal pressures facing 0 Michael Cohen are growing in a wide - ranging investigation of 0 his personal business affairs and 0 his work on behalf of 1 0 his former client , President Trump . In addition to 0 his work for 1 Mr. Trump , 0 he pursued 0 his own business interests , including ventures in real estate , personal loans and investments in taxi medallions .

Figure 6: Co-reference resolution: linking mentions to entities. Bag-of-words representations destroy this structure entirely.

This course is about relating documents to metadata

- ▶ Applied NLP: the documents are rarely interesting on their own.
- ▶ We want to relate **text** to **metadata**.
- ▶ e.g., positive–negative sentiment Y in judicial opinions is not meaningful by itself.
- ▶ But sentiment Y_{ijt} in opinion i , by judge j , at time t :
 - ▶ how does it vary over time t ?
 - ▶ does a judge from party p_j express more negative sentiment toward defendants from group g_i ?

Rule of thumb

A text measure becomes economics only when it is indexed by a unit, a time, and a source of variation.

What counts as a document?

The unit of analysis — the “document” — depends on your question.

- ▶ Fine enough to match the relevant metadata variation.
- ▶ Not finer: that adds dimensionality without adding empirical variation.

What should the document be in each of these cases?

1. Predicting whether a judge is right- or left-wing from written opinions.
2. Testing whether parliamentary speeches become more emotive before an election.
3. Measuring whether newspapers use different sentiment toward different groups.

Dictionary methods: still a workhorse

- ▶ A pre-selected list of words or phrases, counted across documents.
 - ▶ e.g., how often a judge says “justice” versus “efficiency”.
- ▶ General-purpose dictionaries: WordNet, LIWC, moral foundations, and emotion lexicons (Mohammad and Turney 2011; Warriner et al. 2013).
- ▶ Why economists keep using them despite obvious limitations:
 - ▶ transparent and reproducible;
 - ▶ the researcher explicitly “regularizes out” confounding language dimensions rather than letting a model absorb them.
- ▶ Limitation: brittle to negation, domain shift, and sarcasm — and the word list is itself a researcher degree of freedom.

Dictionary methods in practice: policy uncertainty

Baker, Bloom, and Davis (QJE 2016)

For each newspaper on each day since 1985, count articles that contain:

1. “uncertain” OR “uncertainty”; **and**
2. “economic” OR “economy”; **and**
3. “congress” OR “deficit” OR “federal reserve” OR “legislation” OR “regulation” OR “white house”.

Normalize by total articles that month.

But see Keith et al. (2020) on the fragility of this measure:

<https://arxiv.org/abs/2010.04706>

Figure 2: News-Based Economic Policy Uncertainty Index (Jan 1985 – Mar 2013)

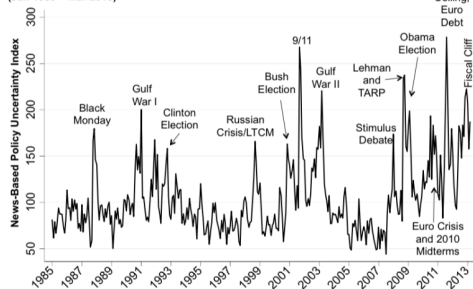


Figure 7: Economic policy uncertainty index.

Supervised measures and their confounders

- ▶ Off-the-shelf sentiment models fail on domain text: in a court opinion, “murder” scores negative regardless of the judge’s tone.
- ▶ More generally:
 - ▶ **supervised** models learn features correlated with the annotated label, not only its true cause;
 - ▶ **unsupervised** models learn correlations between topics and contexts;
 - ▶ **LLMs** inherit both problems, and memorize more language confounders besides.

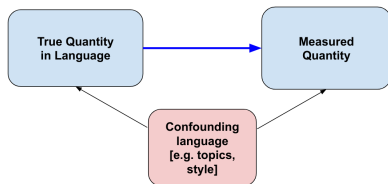


Figure 8: “Bias” in NLP systems is statistical bias.

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From measurement tool to research collaborator

- ▶ First-generation use: LLM as an **annotator** — classify, extract, summarize.
- ▶ Second-generation use: LLM as an **agent** — read a paper, write code, run the analysis, inspect the output, revise.
- ▶ This changes which parts of the research process are scarce.
 - ▶ Cheap: implementation, literature search, robustness checks.
 - ▶ Still scarce: question selection, identification, validation, and judgment about what is interesting.
- ▶ Disclosure norms are forming quickly — substantive AI use is increasingly expected to be reported, like any other research tool.

AI for literature review

- ▶ Retrieval-augmented tools ground answers in a fixed corpus of papers you supply, which sharply reduces (but does not eliminate) fabrication.
- ▶ Useful patterns:
 - ▶ build a corpus of 50–200 PDFs on your topic and query *that*, rather than the open web;
 - ▶ ask for *disagreements* across papers, not summaries;
 - ▶ ask for each paper's identification strategy in one sentence.
- ▶ Failure modes: confident citation of non-existent work; over-weighting of highly cited older papers; silent omission of non-English literatures.

Non-negotiable

Every citation that survives into your draft must have been opened and read.

“Prompt engineering” for research

Prompting Science Report 1: Prompt Engineering is Complicated and Contingent

10 Pages

Posted: 5 May 2025

[Lennart Meincke](#)

University of Pennsylvania; The Wharton School; WHU - Otto Beisheim School of Management

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Date Written: March 04, 2025

Abstract

This is the first of a series of short reports that seek to help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. In this report, we demonstrate two things:

- There is no single standard for measuring whether a Large Language Model (LLM) passes a benchmark, and that choosing a standard has a big impact on how well the LLM does on that benchmark. The standard you choose will depend on your goals for using an LLM in a particular case.

- It is hard to know in advance whether a particular prompting approach will help or harm the LLM's ability to answer any particular question. Specifically, we find that sometimes being polite to the LLM helps performance, and sometimes it lowers performance. We also find that constraining the AI's answers helps performance in some cases, though it may lower performance in other cases.

Figure 9: Prompt structure for annotation tasks.

- ▶ Treat the prompt as a **codebook**: written so that a human coder and the model would produce the same label.

Figure 10: Iterating on instructions with a labeled development set.

Using LLM outputs in regressions

Using Large Language Model Annotations for the Social Sciences:
A General Framework of Using Predicted Variables
in Downstream Analyses*

Naoki Egami[†] Musashi Hinck[‡] Brandon M. Stewart[§] Hanying Wei[¶]

First Version: May 13, 2024
This Version: November 17, 2024

Abstract

Social scientists use automated annotation methods, such as supervised machine learning and, more recently, large language models (LLMs), that can predict labels and generate text-based variables. While such predicted text-based variables are often analyzed as if they were observed without errors, we show that ignoring prediction errors in the automated annotation step leads to substantial bias and invalid confidence intervals in downstream analyses, even if the accuracy of the automated annotations is high, e.g., above 90%. We propose a framework of *design-based supervised learning* (DSL) that can provide valid statistical estimates, even when predicted variables contain non-random prediction errors. DSL employs a doubly robust procedure to combine predicted labels and a smaller number of expert annotations. DSL allows scholars to apply advances in LLMs to social science research while maintaining statistical validity. We illustrate its general applicability using two applications where the outcome and independent variables are text-based.

Figure 11: Errors in LLM-generated variables.

- Practical minimum: hand-label a validation sample, report accuracy *by subgroup*, and show how the estimate moves under a measurement-error correction.

Inference for Regression with Variables Generated by AI
or Machine Learning*

Laura Battaglia[†] Timothy Christensen[‡]
Stephen Hansen[§] Szymon Sacher[¶]

April 30, 2025

Abstract

Researchers now routinely use AI or other machine learning methods to estimate latent variables of economic interest, then plug-in the estimates as covariates in a regression. We show both theoretically and empirically that naively treating AI/ML-generated variables as “data” leads to biased estimates and invalid inference. To restore valid inference, we propose two methods: (1) an explicit bias correction with bias-corrected confidence intervals, and (2) joint estimation of the regression parameters and latent variables. We illustrate these ideas through applications involving label imputation, dimensionality reduction, and index construction via classification and aggregation.

Figure 12: Consequences for downstream estimates.

Case study: can agents replicate published research?

Kohler et al., “Read the Paper, Write the Code”

- ▶ AI models can now write and execute code autonomously.
- ▶ Social science journals increasingly mandate replication packages (data + code) — yet these packages are rarely re-run, and the reproducibility crisis persists.

Research question

Can AI agents automatically replicate empirical social science research, given only the paper and the raw data?

- ▶ Agents are evaluated against human-verified replications from the Institute for Replication (I4R).

What level of replication?

Coding-level replication:

- Given:** Paper (methods description)
+ Raw data
- Held out:** Original code
Original results
- Task:** Reproduce all tables and figures

This is *not* conceptual replication with new data or new methods.

A finished research product has three components:

- ▶ **Inputs** (data)
- ▶ **Specifications** (methods)
- ▶ **Outputs** (results)

Key question: is the paper's verbal description of methods sufficient for an agent to write new code that reproduces the original results?

The replication pipeline

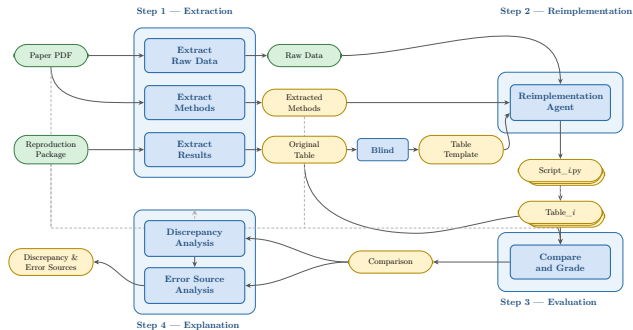


Figure 13: The replicator never sees the original paper, results, or code.

Extractor: methodology only, structured output. **Results extractor:** original cell values, then blinded templates. **Replicator:** fills in each cell from raw data. **Judge:** deterministic cell-by-cell grading (A–F). **Explainer:** root-cause diagnosis with full access.

Benchmark and design

Ground truth (I4R):

- ▶ 110 papers independently replicated by trained researchers;
- ▶ 46 with verified *perfect* reproduction and sufficient raw data form the benchmark;
- ▶ $\Rightarrow$ any discrepancy is attributable to the **agent**, not the paper.

Contamination checks:

- ▶ split papers by model knowledge cutoff;
- ▶ guardrail audit on file paths and URLs accessed;
- ▶ hardcoding audit for results written as numeric literals.

Pre- versus post-cutoff performance differs insignificantly — memorization is not driving the results.

Results: agents get the sign, struggle with the digits

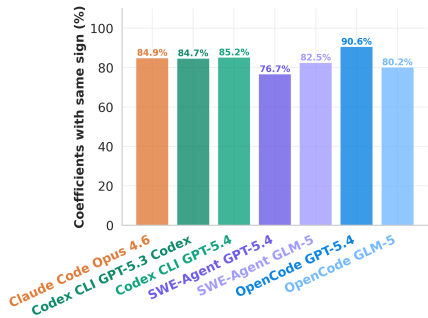


Figure 14: Share of reproduced coefficients with the same sign as the original.

Best agent: correct sign more than 90% of the time, and more than 80% of coefficients inside the original 95% CI. Even the weakest agent matches sign more than 77% of the time. Exact full-paper reproduction remains hard.

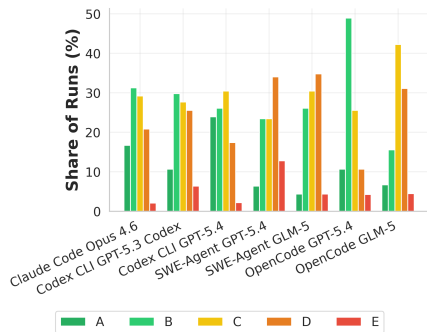


Figure 15: Distribution of paper-level grades by agent.

Where do the errors come from?

Largest source: paper underspecification

- ▶ 50% of errors are paper–code conflicts: the paper does not describe the method precisely enough to write the code;
- ▶ a further 16% are missing data;
- ▶ $\Rightarrow$ about two thirds of errors are **human**, not AI.

For the best agent, genuine agent mistakes account for only about 8% of discrepancies.

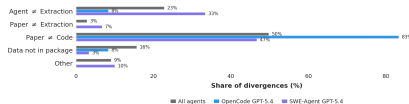


Figure 16: Root causes of discrepancies, by agent.

Implications for how we do research

For social science practice:

- ▶ methods sections alone are frequently *insufficient* to reproduce an analysis;
- ▶ structured, machine-readable methods summaries may become part of submission;
- ▶ automated reproducibility checks are now technically feasible.

For AI agents:

- ▶ scaffold–model alignment matters as much as raw model quality;
- ▶ the frontier moves fast enough that results date within months;
- ▶ error rates would fall substantially with better *paper* practices.

Direction of travel

A first step toward autonomous *conceptual* reproduction — from data collection through method design.

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Data/Compute 1.0 → AI 1.0 → Data/Compute 2.0

This generation of AI was built with pre-AI data, compute, ML, and values.

What will these look like in a post-AI era?

- ▶ **Data:** AI-boosted digitization (OCR, transcription, translation, sensors); RAG-optimized vector stores; AI-generated data.
- ▶ **Compute:** processors designed for inference rather than graphics; AI-optimized chipsets.
- ▶ **ML:** AI-optimized codebases and programming workflows; architectures beyond the transformer.
- ▶ **Values / alignment:** integration of metadata, causal modeling, and behavioral responses into the objective.

Reasoning models and verifiable rewards

- ▶ Chain-of-thought training plus reinforcement learning against *verifiable* rewards (RLVR) rather than human preferences (RLHF).
- ▶ Open-weight reasoning models made frontier-adjacent capability cheap and locally runnable.
- ▶ Economic implication: the marginal cost of a careful analytical step is falling faster than the cost of a creative one.
- ▶ Open question: RLHF is probabilistic and encourages interpolation; RLVR is discrete and discourages it. What does that do to the model's worldview?

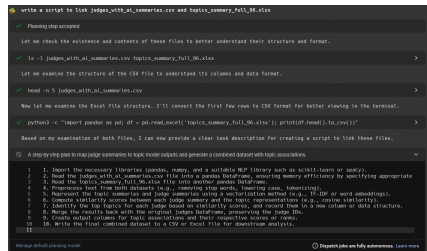
Metric	DeepSeek V3	Llama 3.1
Parameters	671B total (37B active per token)	405B
GPU Type	NVIDIA H800	NVIDIA H100
GPU Count	2,048	Up to 16,000
Training Duration	~2 months	~2.6 months (estimated)
Tokens Processed	14.8T	15.6T
GPU Hours	2.788M	~30.8M
Training Cost	~\$5.6M	~\$2.4M–123.2M (estimated)
Cost per Trillion Tokens	~\$378K	~5.93M–7.90M

Note: Cost estimations uses an average of \$2/hour for H800 GPUs (DeepSeek V3) and \$3/hour for H100 GPUs (Llama 3.1) based on rental GPU prices.

Figure 17: Open-weight reasoning models compressed the capability–cost frontier.

Agents and orchestration

- ▶ An agent = model + tools + a loop that decides when to stop.
- ▶ Research-relevant capabilities: file system access, code execution, web retrieval, long-horizon planning, parallel sub-agents.
- ▶ What this enables: running twenty specification variants overnight; auditing a co-author's code; scraping and cleaning a corpus end to end.
- ▶ What it does not enable: knowing whether the specification was the right one.



```
write a script to link judges_with_at_summaries.csv and topics_summary_full_96.xlsx

Planning step accepted

Let me check the existence and contents of these files to better understand their structure and format.

ls -l judges_with_at_summaries.csv topics_summary_full_96.xlsx

Let me examine the structure of the CSV file to understand its columns and data format.

head -n 5 judges_with_at_summaries.csv

Now let me examine the Excel file structure. I'll convert the first few rows to CSV format for better viewing in the terminal.

python3 -c "import pandas as pd; df = pd.read_excel('topics_summary_full_96.xlsx'); print(df.head().to_csv())"

Based on my examination of both files, I can now provide a clear task description for creating a script to link these files.

A step-by-step plan to map judge summaries to topic model outputs and generate a combined dataset with topic associations.

1. Import the necessary libraries (pandas, numpy, and a suitable NLP library such as sklearn, spacy, or spaCy).
2. Read the judges_with_at_summaries.csv file into a pandas DataFrame, ensuring memory efficiency by specifying appropriate chunksize.
3. Read the topics_summary_full_96.xlsx file into another pandas DataFrame.
4. Preprocess text from both datasets (e.g., removing stop words, lowering case, tokenizing).
5. Represent the topic summaries and judge summaries using a vectorization method (e.g., TF-IDF or word embeddings).
6. Compute similarity scores between each judge summary and the topic representations (e.g., cosine similarity).
7. Identify the top topics for each judge based on similarity scores, and record them in a new column or data structure.
8. Merge the results back with the original judge DataFrame, preserving the judge ID.
9. Create output columns for topic associations and their respective scores or ranks.
10. Write the final combined dataset to a CSV or Excel file for downstream analysis.

11
```

Figure 18: A coding agent operating directly in a research repository.

Multimodality: economics beyond text

- ▶ Images: satellite data for growth and conflict; street-level imagery for neighborhood characteristics; scanned historical documents.
- ▶ Video and audio: parliamentary and courtroom recordings, campaign ads, earnings calls — prosody and affect are measurable, not just words.
- ▶ Joint text–image models allow measures that neither modality supports alone (e.g., visual framing in political news).
- ▶ These carry the *same* measurement-error problems as text, plus much weaker validation traditions.

Practical note

Image and video pipelines come later in the Summer School; the econometric lessons transfer directly from the text case.

The policy problems will not stay outside the seminar room

- ▶ **Accuracy:** hallucination, silent failure, unreproducible outputs.
- ▶ **Equity:** differential error rates across groups; asymmetric costs of those errors.
- ▶ **Externalities:** privacy; information advantage held by the model provider over the user.
- ▶ **Behavioral responses by subjects:** gaming measurable features; under-investing in the skills the system now supplies.
- ▶ **Behavioral responses by decision-makers:** ignoring the model because it is a black box — or deferring to it and skipping due diligence.

AI and human capital

- ▶ AI can *reduce the cost of learning*: an always-available tutor that explains concepts on demand.
- ▶ AI can also *substitute for the learning process*: the student completes the task without acquiring the skill.
- ▶ If students are shortsighted, substitution blocks the foundations needed for advanced work — the “de-skilling” problem.
- ▶ Market response: take-home work loses signal value; weight shifts to live interviews, proctored assessment, and demonstrated judgment.

Relevance to this course

We assume you use AI throughout. The skill we care about is not “can you write the code” but “can you tell when the output is wrong”.

Open ethical and political questions

- ▶ What does it mean to delegate legal interpretation and judicial decision making to machines?
- ▶ What are the long-run implications for a system's adaptiveness to change?
 - ▶ political and cultural effects;
 - ▶ effects on trust in institutions and on the incentive to appeal.
- ▶ Who chooses what the model values, and by what procedure?
- ▶ Economists have comparative advantage here: incentives, welfare, equilibrium responses, and measurement.

What do human researchers still do?

- ▶ **Choose the question.** Models optimize objectives; they do not decide what is worth knowing.
- ▶ **Own the identifying assumption.** No amount of generated text makes an exclusion restriction credible.
- ▶ **Validate the measure** against ground truth the model has never seen.
- ▶ **Take responsibility.** Authorship means accountability for every number in the paper.

The bet behind this Summer School

The returns to economic judgment go *up*, not down, as implementation becomes free — provided you understand the tools well enough to supervise them.

Where we go from here

- ▶ Next, today: machine learning foundations, embeddings and clustering, then text and image data.
- ▶ Tuesday: econometrics with machine-generated variables; then how language models work and what alignment does to them.
- ▶ Wednesday: working with agents; building research software with AI.
- ▶ Thursday: training and adapting LLMs in practice.
- ▶ Week 2: faculty research sessions and your own project.

Questions and comments?